

SolarGuard: An End-to-End Decision Support System Linking Unsupervised Anomaly Detection to Inventory and Financial Optimization in PV Operations

Abstract. In import-dependent photovoltaic markets like Vietnam, existing research addresses maintenance in isolation: machine learning studies focus strictly on fault classification while neglecting supply-chain logistics, whereas financial frameworks assume immediate spare-part availability—an assumption rendered invalid by 14-day procurement lead times. To bridge this gap, this paper proposes SolarGuard, an end-to-end, data-centric decision-support framework that translates standard 15-minute telemetry into inventory and financial decision-support strategies. The first layer addresses the scarcity of labeled fault data via a zero-shot, unsupervised pipeline combining STL decomposition and Isolation Forests. Evaluated through a domain-specific asymmetric Expected Maintenance Cost evaluation on a 26.15 MWp commercial site (219 inverters), the pipeline achieves a 98.02% recall rate ($F1=0.7046$), substantially reducing the alert volume observed in the evaluated supervised baselines. The inventory layer translates these risk signals into a deterministic stocking policy, estimated to preserve 376.65 MWh of generation under the evaluated scenario. Finally, integrating this avoided downtime into the project's P90 financial model yields an estimated NPV gain of +USD 63.5k based on annualized pilot extrapolation. Ultimately, this framework provides a traceable decision-support pathway from raw technical signals to investment-relevant decisions, aligning technical operations with stringent mandates like Vietnam's Decree 135/2024 and DPPA.

Keywords: rooftop solar PV; predictive maintenance; anomaly detection; inverter inventory optimization; performance ratio; financial decision-making.

1 Introduction

Driven by feed-in tariffs, Vietnam's rooftop solar (RTS) sector expanded to 9.3 GW by 2020 [1], [2], [3]. However, predominantly reactive O&M practices among non-technical owners induce an estimated 8% yield underperformance [4]. Inverter faults constitute the dominant failure mode, driving 76% of outages and up to 36% of generation losses [5], [6]. Unlike gradual soiling, these faults cause immediate asset failure, and replacement within Vietnam's import-dependent supply chain often spans weeks, severely penalizing long-term yields [7]. Yet, conventional asset valuation models treat financial returns as fixed, ignoring the critical feedback loop between maintenance efficacy and investment profitability [8].

Predictive maintenance can cut O&M cost by 35% and add 9% uptime [4]. However, existing literature addresses anomaly detection [9], [10] and economic modeling [11], [12] in isolation. No prior framework closes the operational loop from AI-generated alerts to spare-part logistics and investor-level metrics (NPV, IRR), particularly within label-scarce commercial environments. This paper bridges this gap with a three-layer framework - anomaly detection, inventory optimization, and financial assessment -

validated on an empirical 26.15 MWp Vietnamese RTS portfolio under tropical monsoon conditions.

The main contributions are threefold: (1) an integrated pipeline linking unsupervised anomaly alerts directly to deterministic inventory routing, integrating technical telemetry with supply-chain logistics; (2) a zero-shot, deployment-oriented triage layer combining STL decomposition and Isolation Forests to reduce operational false alarms, alert volume, and operator workload; and (3) a techno-economic module mapping mitigated downtime into an existing project-level P90 financial model to quantify NPV and IRR preservation under evolving regulatory frameworks.

2 Related Work

2.1 PV O&M and Predictive Maintenance

Effective operation and maintenance (O&M) is widely recognised as a critical determinant of photovoltaic (PV) system lifetime performance. Predictive maintenance (PdM), defined as the use of data-driven models to anticipate equipment failures before they occur and trigger timely intervention, represents the most advanced O&M strategy on the maintenance spectrum [13]. Despite its potential, prior studies have identified a pervasive “install-and-forget” mentality in the commercial rooftop sector, resulting in substantial energy yield losses, and found that no third-party PdM tool has reached widespread commercial deployment [13]. Predictive analytics also remains underexplored relative to corrective and preventive strategies [5], while integrated IoT-AI PdM frameworks for small-scale commercial rooftop systems in emerging markets remain limited [14]. Critically, commercial rooftop PV systems differ from utility-scale plants in that they are more fragmented, less instrumented, and often operated without dedicated O&M teams, making the direct transfer of large-plant maintenance frameworks to this context problematic [13], [14].

2.2 AI-Based Fault Detection and Anomaly Detection in PV Systems

AI-based fault detection has evolved from broad taxonomies toward empirical deployment, yet existing pipelines consistently terminate at fault classification without connecting to O&M cost implications or spare-part readiness [15], [16], [17]. Four algorithmic approaches emerge from the literature:

Table 1. Summary Table of Related Work

Approach	Strength	Limitation for This Study
Supervised fault classification	High accuracy (up to 98.7%)	Requires labelled failure records, rarely available in commercial rooftop contexts
Residual-based anomaly detection	Label-free; suits irradiance + yield data	No connection to inventory or financial layer
Unsupervised (Isolation Forest / Autoencoder)	Label-free; scales to multi-inverter fleets	Stops at classification output

Sensor-heavy classification	Fine-grained fault taxonomy	Requires dense instrumentation unavailable in most rooftop deployments
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2.3 Maintenance Cost Optimisation and Financial Evaluation

The economic optimisation of PV maintenance has developed more recently than technical fault-detection research. Earlier work mainly compared corrective and preventive maintenance strategies before the rise of ML-based PdM [12]. Recent prescriptive studies show that maintenance scheduling optimisation can reduce costs [18], [19], but they rarely integrate AI-based fault signals into inventory decisions or investor-level financial metrics. From a project-finance perspective, PdM should be evaluated not only by detection accuracy but also by its ability to reduce downtime, preserve cash flow, and improve NPV and IRR, as O&M cost is a key driver of rooftop PV project IRR in Southeast Asia [20]. DSCR is left to future work, as it requires a full debt schedule. In Vietnam, long inverter procurement lead times mean that a PdM alert has limited value unless replacement units are available. The key gap is therefore the link between prediction, spare-part readiness, and financial performance.

3 Case Study and Dataset

3.1 Project and Dataset Overview

The case study is an anonymized commercial rooftop solar PV project in Vietnam with an installed capacity of 26.15 MWp and $N=219$ physical inverters. The project couples high-frequency inverter monitoring with a concrete O&M problem: when an inverter becomes unavailable and no replacement unit is locally ready, the operator incurs generation loss during the procurement lead time. For confidentiality, the project name, exact coordinates, owner, and developer are anonymized. The inverter count $N=219$ is used consistently across dataset statistics, lost-capacity estimation, the inventory layer, and the financial layer. When inverter-level capacity ratings become available, the model should use each inverter's connected capacity C_i rather than the project-level average $\bar{C} = 26,150/219 \approx 119.4$ kW.

Table 2. Case Study Project Information And Dataset Statistics

Item	Value/description
Installed capacity	26.15 MWp (fixed project-level value)
System type	Commercial rooftop solar PV
Location	Vietnam; name and coordinates anonymized
Physical inverters N	219
Raw data frequency	15-minute monitoring-platform logs

Observed period	15 months, from 2025-03-01 to 2026-05-31
Modelling unit	one inverter $\times$ one day (inverter-day)
Primary computed metric	daily PR and peer-relative PR
Fault labels	Candidate signals from triage process; pending ground-truth validation
Avg. lost capacity per inverter	$\bar{C} \approx 119.4 \text{ kW}$, used only when inverter-level C_i is unavailable

3.2 Data Sources and Variables

The dataset has two raw streams: inverter-level monitoring (timestamped energy readings at 15-minute resolution for $N = 219$ inverters) and irradiance (plane-of-array irradiance in W/m^2 , converted to kWh/m^2 per 15-minute interval for PR normalization). The modelling unit is the inverter-day. From these streams we derive four core variables-daily energy yield, daily POA irradiation, inverter installed capacity, and a timestamped inverter identifier-used to construct daily PR and peer-relative PR as inputs to the anomaly layer.

Table 3. Data sources and modelling variables

Data source / variable	Definition	Use in framework
Inverter generation records	Timestamped energy per inverter at 15-min intervals (or recovered from cumulative counters)	Raw input for daily yield and PR
Irradiation measurements	POA / site irradiance; W/m^2 converted to kWh/m^2 per interval	Normalization input for daily PR
Installed / connected capacity C_i	Rated or connected DC capacity at inverter level	Denominator in PR and lost-capacity
Timestamp and inverter ID	Date-time and device identifier after standardization	Panel construction and rolling scoring
Financial assumptions	P90 FM cashflow, electricity price, discount rate, FX	Layer 3 financial recalculation
Inventory assumptions	Lead time, install time, current spare stock, unit cost, holding rate	Layer 2 inventory engine

3.3 Preprocessing Pipeline

Let X_{raw} denote the raw monitoring logs. Preprocessing is the composition of five operators that produce the clean inverter-day panel P :

$$P = \Pi(X_{\text{raw}}) = (\phi \circ \mu \circ \nu \circ \kappa \circ \sigma)(X_{\text{raw}}),$$

where σ standardizes identifiers/timestamps, κ recovers interval energy, ν applies physical-validity masking, μ aggregates to inverter-day, and ϕ builds rolling features. The pipeline records intermediate transformations and quality flags to support reproducibility and preserves an audit trail through explicit quality flags rather than silent deletion.

Energy recovery (κ). Where the platform exports cumulative energy $E_{i,t}$, interval energy is recovered by first-differencing within each inverter-day, clipping counter resets to zero:

$$e_{i,t} = \max(E_{i,t} - E_{i,t-1}, 0).$$

Irradiance conversion. POA irradiance G_t (W/m^2) is converted to interval energy with $\Delta t = 0.25$ h:

$$H_t = \frac{G_t \cdot \Delta t}{1000} \quad (kWh/m^2).$$

Daily aggregation (μ). For inverter i on day d over intervals $t \in T_d$:

$$Y_{i,d} = \sum_{t \in T_d} e_{i,t}, \quad H_d = \sum_{t \in T_d} H_t.$$

Performance ratio and peer-relative PR. With installed capacity C_i and fleet I :

$$PR_{i,d} = \frac{Y_{i,d}}{H_d \cdot C_i}, \quad \tilde{PR}_{i,d} = \frac{PR_{i,d}}{\text{median}_{j \in I} PR_{j,d}}.$$

Validity mask (ν). Each inverter-day carries a quality flag $m_{i,d} \in \{0, 1\}$; flagged rows are retained for audit but excluded from scoring:

$$m_{i,d} = 1 \left[Y_{i,d} \geq 0 \wedge H_d > 0 \wedge C_i > 0 \wedge PR_{i,d} \in [0, 1.2] \right].$$

Rolling features (ϕ). For window $w \in \{14, 30\}$ days the feature map summarizes the recent panel into the scoring vector $x_{i,d}$:

$$x_{i,d} = \phi(\{PR_{i,\tau}, \tilde{PR}_{i,\tau}, \delta_{i,\tau}\}_{\tau=d-w+1}^d), \quad \delta_{i,d} = \max(1 - \tilde{PR}_{i,d}, 0).$$

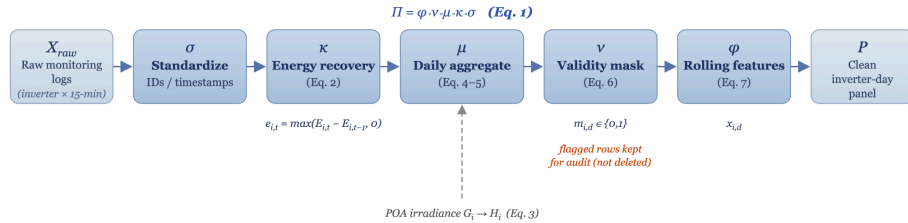


Fig. 1. Data-preprocessing pipeline for inverter-day panel construction and audit retention.

3.4 Dataset Structure

The processed data are organized at two analytical levels: an inverter-day panel for anomaly scoring, and a monthly aggregation table for inventory and financial mapping. The monthly table summarizes daily signals into decision variables (candidate events, expected demand, recommended stock, generation saved); it does not replace daily scoring.

Table 4. Final dataset structure

Level	Unit	Key fields
Raw monitoring	inverter $\times$ timestamp	energy, irradiance, ID
Daily panel	inverter $\times$ day	$Y, H, C, PR, \tilde{PR}, \delta, m$
Daily risk output	inverter $\times$ date	$s_{i,d}$, class $c_{i,d}$, rolling indicators, candidate flag
Monthly decision	month	candidate events, λ, S^* , generation saved
Quarterly FM mapping	FM quarter	production saved, revenue delta, net cashflow with AI

4 Methodology: Three-Layer AI O&M Decision Framework

4.1 Problem Formulation

The framework does not treat AI as a stand-alone fault classifier. Instead, the entire system is a composition of three maps that transform raw operating data into investor-relevant financial deltas:

$$X_{raw} \rightarrow^{\Pi} P \rightarrow^{f_1} r \rightarrow^{f_2} u \rightarrow^{f_3} \theta,$$

where Π is preprocessing (Eq. 1), f_1 the anomaly/risk map producing risk vector $r = \{s_{i,d}\}$, f_2 the inventory-decision map producing the reorder decision $u = (S^*, Q)$, and f_3 the financial map producing $\theta = (IRR, NPV)$. Indices are inverters $i \in I = \{1, \dots, N\}$, days $d \in D$, with lead-time window L (China procurement + installation).

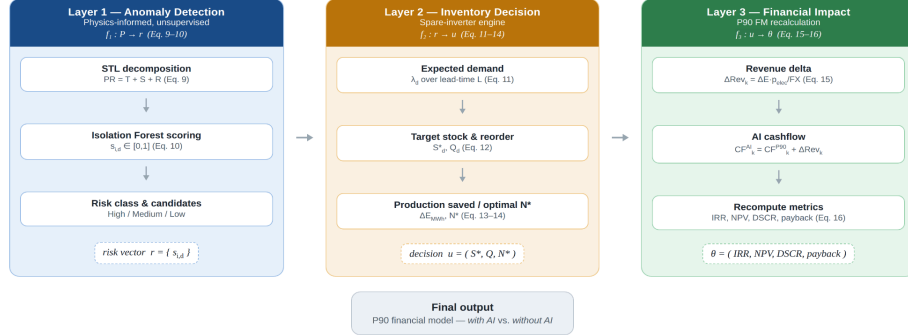


Fig. 2. Three-layer AI O&M decision framework linking anomaly detection, inventory planning, and financial evaluation.

4.2 Layer 1 - Anomaly Detection and Risk Classification

Layer 1 is designed for label-scarce field conditions, combining physics-informed normalization, peer comparison, and unsupervised scoring. Peer-relative PR (Eq. 5) reduces the likelihood that fleet-wide weather variation is mistaken for inverter-specific loss. STL decomposition separates normal temporal structure from abnormal residuals:

$$PR_{i,d} = T_{i,d} + S_{i,d} + R_{i,d}.$$

The feature vector $x_{i,d}$ (Eq. 7) is scored by an Isolation Forest, appropriate because confirmed labels are unavailable and the objective is to *rank* unusual behavior rather than train a supervised classifier. The score is mapped to an operational risk class with thresholds $\tau_M < \tau_H$:

$$s_{i,d} = IF(x_{i,d}) \in [0, 1], \quad c_{i,d} = \{ \text{High } s_{i,d} \geq \tau_H, \text{ Medium } \tau_M \leq s_{i,d} < \tau_H, \text{ Low } s_{i,d} < \tau_M \}.$$

These classes are decision-support labels, not confirmed hardware faults.

Table 5. Rolling features used for risk scoring.

Feature	Definition	Reason
Mean PR 14d/30d	rolling mean of $PR_{i,d}$	sustained underperformance
Mean rel. PR 14d/30d	rolling mean of $\tilde{PR}_{i,d}$	separates device loss from weather
Peer deficit $\delta_{i,d}$	$\max(1 - \tilde{PR}_{i,d}, 0)$	non-negative severity signal
Severe-/moderate-low rate	share of days below thresholds	persistent vs. early-warning loss
Trend decline 30d	negative slope of $\tilde{PR}$	gradual degradation
Latest peer deficit	most recent $\delta_{i,d}$	responsiveness to recent change

4.3 Layer 2 - Inventory Decision Engine

Layer 2 converts risk into a spare-inverter decision over the lead-time window L . Importantly, the anomaly warning does not eliminate the 14-day international procurement lead time. Instead, the inventory policy maintains a local reserve of spare inverters, while the warning signal supports spare reservation, dispatch planning, and replacement scheduling. Once a locally stocked spare is consumed, the reorder policy replenishes inventory according to the target stock level defined below. Expected failure demand aggregates a risk-to-replacement map $g(\cdot)$ over the fleet:

$$\lambda_d = \sum_{i \in I} Pr(replace_i \leq d + L) \approx \sum_{i \in I} g(s_{i,d}).$$

With inventory position $IP_d = q_d^{on-hand} + q_d^{on-order} - q_d^{reserved}$, safety factor z_α , and demand spread σ_λ , the target stock and reorder quantity are

$$S_d^* = \min(S_{max}, \max(S_{safety}, [\lambda_d + z_\alpha \sigma_\lambda])), \quad Q_d = \max(0, S_d^* - IP_d).$$

The engine compares three policies: reactive procurement (order after failure; downtime $\approx L + \tau_{inst}$), the current fixed pre-stock policy, and the proposed risk-informed reorder policy. The operational delta mapped to finance is **production saved**, computed from avoided downtime $\Delta_{down} = D_{base} - D_{AI}$:

$\Delta E_{MWh} = \frac{\Delta_{down} \cdot \bar{C} \cdot h \cdot n_{aff}}{1000}$, with $\bar{C}$ the lost capacity per inverter, h generation hours/day, and n_{aff} affected inverters. The inventory-cost-minimizing spare count under the stated assumptions balances expected downtime cost against holding cost:

$$N^* = \arg \min_N [c_{loss} \cdot E[shortfall(N)] + c_{hold} \cdot N \cdot V_{inv}].$$

4.4 Layer 3 - Financial Impact and P90 FM Recalculation

Layer 3 maps the production delta into the existing P90 financial model without redefining CAPEX, debt, tax, baseline OPEX, or tariff logic - only the avoided production loss changes. With electricity price p_{elec} and exchange rate FX , the period revenue delta and AI cashflow are

$$\Delta Rev_k = \frac{\Delta E_{MWh,k} \cdot p_{elec}}{FX}, \quad CF_k^{AI} = CF_k^{P90} + \Delta Rev_k.$$

The P90 FM then recomputes the metrics on CF^{AI} , e.g.

$$NPV^{AI} = \sum_k \frac{CF_k^{AI}}{(1+r)^k}, \quad \sum_k \frac{CF_k^{AI}}{(1+IRR^{AI})^k} = 0.$$

Results are reported as *P90 without AI* vs. *P90 with AI*. When data end at a cutoff, future savings are scenario-based forecasts, not realized AI predictions - a distinction that matters for long-horizon projections.

Table 6. Financial mapping logic

Item	Without AI (P90 FM)	With AI mapping
Production loss	baseline waiting loss	reduced by ΔE_{MWh}
CAPEX / OPEX / debt / tax	fixed	unchanged
Net cashflow	CF_k^{P90}	$CF_k^{P90} + \Delta Rev_k$
IRR / NPV	from P90 FM	recomputed (Eq. 16)

4.5 Adaptive Mechanism (No-Retrain Operation)

Each run updates without retraining: append data $\rightarrow$ recompute rolling features (Eq. 7) $\rightarrow$ update scores (Eq. 10) $\rightarrow$ refresh demand (Eq. 11) and stock (Eq. 12) $\rightarrow$ refresh FM mapping (Eq. 15–16). If confirmed maintenance/replacement labels later arrive, they are joined to past risk outputs to validate precision, recall, F_1 , and lead time before failure; until then $s_{i,d}$ is a triage signal, not a calibrated failure probability.

5 Experimental Results

This section validates the three-layer AI O&M framework on the anonymized 26.15 MWp rooftop PV case study (N = 219 inverters). Because confirmed replacement labels are sparse, the system is evaluated as a hybrid maintenance-triage pipeline rather than a standalone classifier, distinguishing hard binary decisions from ranked watchlist behavior.

5.1 Experimental Setup and Cost-Sensitive Evaluation

This section validates the three-layer decision-support framework on the anonymized 26.15 MWp rooftop PV case study. Because naturally occurring replacement labels are inherently sparse, we evaluate the framework on a hybrid suite comprising 10 documented real-world inverter failures and a controlled synthetic fault injection protocol. Specifically, progressive degradation was simulated by applying a linear decay factor (0.5% to 5% per day, derived from documented degradation profiles) to the interval yield over a 14-day horizon, whereas abrupt faults were modeled as instantaneous capacity drops (20% to 100%), with parameters sampled from historical fault distributions.

To rigorously expand the dataset, the synthetic protocol introduces both progressive degradation and abrupt hard faults grounded in documented failure modes [5], [6], [13]. To ensure ecological validity and prevent model bias, injections are strictly

applied to sunny-day regimes (irradiance ≥ 0.5 kWh/m²) and calibrated against same-day peer behavior. This ensures that the injected anomalies represent genuine hardware deviations rather than weather artifacts.

Furthermore, since commercial maintenance is fundamentally an economic tradeoff rather than a pure classification task, we introduce an Expected Maintenance Cost (EMC) metric. This model applies an asymmetric financial penalty to False Positives (representing wasted inspection labor) versus False Negatives (representing severe unmitigated generation loss). The detailed dataset scale, fault injection parameters, cost assumptions, and the supervised Gradient Boosting baselines used to benchmark the proposed zero-shot framework are summarized in Table 7.

Table 7. Experimental setup and benchmark configuration

Feature / Metric	Data Details
Data Scale	93,277 rows across 219 devices
Timeline	2025-03-01 to 2026-05-31 (2/3 train, 1/3 test split)
Evaluation Suite	101 target events within the test window (derived from 300 injected synthetic events)
Lead Window	30 days early-warning evaluation
Cost Model	False Positive (FP) = USD 10.00; False Negative (FN) = USD 1,500.00
AI Logic	Configuration: risk_score ≥ 13 and relative PR < 0.85
Zero-Shot Baselines	Proposed SolarGuard, Isolation Forest
Supervised Baselines	XGBoost, LightGBM, HistGradientBoosting

5.2 STL Decomposition

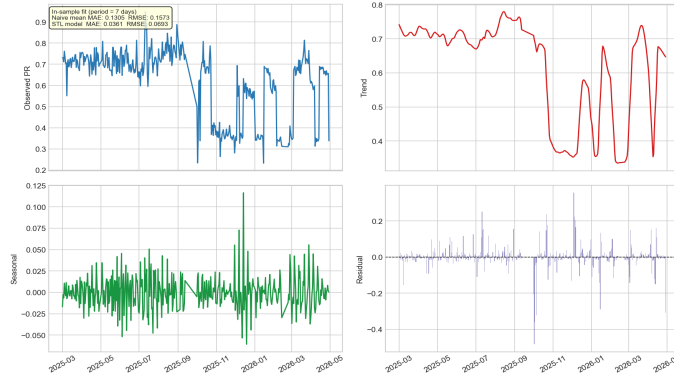


Fig. 3. STL decomposition of daily mean PR (observed, trend, seasonal, residual).

STL decomposition of the fleet daily mean PR (Figure 3) achieves RMSE 0.1866 versus 0.2367 for a naive mean baseline - a 21.15% improvement - capturing a late-2025 trend shift and a weekly seasonal pattern, and supplying the residual features used by Layer 1.

5.3 Anomaly Detection

Supervised baselines are fit on the chronological train window ($\leq 2025-10-26$); probability cutoffs are tuned on the validation window (42 events) to minimize EMC and then frozen for test evaluation (101 events). Despite this cost-sensitive validation tuning, Table 8 shows that supervised models still incur higher test EMC than SolarGuard because aggressive recall on synthetic training labels propagates excess false alerts into the held-out test window.

Table 8. Event-centric detection metrics evaluated over the 101-event test suite, mapped from inverter-day risk scores within a 30-day lead-time window.

Method	Zero-shot	Precision	Recall	F1-Score	Avg early (d)	EMC
SolarGuard	Yes	0.55	0.98	0.70	2.51	\$3,810
Isolation Forest	Yes	0.21	0.99	0.35	6.38	\$5,240
XGBoost	No	0.74	0.92	0.82	11.53	\$6,770
HistGradientBoosting	No	0.62	0.94	0.75	13.36	\$9,590
LightGBM	No	0.56	0.96	0.71	14.50	\$12,320

5.4 Inventory Cost-Benefit Analysis

Layer 2 evaluates three inventory strategies rather than only comparing reactive procurement against AI. This is necessary because the operator’s current practice is already to hold spare inverters on site. The comparison therefore includes reactive procurement ($N = 0$), the current fixed pre-stock policy ($N = 10$), and the AI cost-minimizing stock level ($N = 5$).

Over the five-month pilot window, the pipeline covers 52 replacement-candidate events. Relative to reactive downtime exposure, the AI stocking policy is estimated to preserve 376.65 MWh of generation.

Table 9. Inventory Strategy Cost-Benefit Comparison

Metric	Reactive procurement	Fixed pre-stock: 10 units	AI-driven inventory
Spare-stock logic	0 units; buy after failure	Hold 10 units from start	Inventory-cost-minimizing $N = 5$
Candidate events covered	52	52	52

Metric	Reactive procurement	Fixed pre-stock: 10 units	AI-driven inventory
Generation loss cost	\$31,239.22	\$184.31	\$385.67
Inventory holding cost	\$0	\$980.39	\$490.20
Total inventory cost	\$31,239.22	\$1,164.71	\$875.87
Saving vs reactive	–	\$30,074.51	\$30,363.35

AI monthly service fee / CAPEX is excluded from Table 9 and included in Table 10 to avoid mixing software OPEX with spare-holding cost.

Both stocked policies dominate reactive procurement under the 14-day lead time. On inventory costs alone, AI ($N = 5$) cuts total cost by ~USD 30.4k (−97%) vs reactive and remains USD 289 cheaper than fixed pre-stock ($N = 10$), by avoiding overstock once simultaneous candidate peaks are covered. AI service fees are excluded here and assessed in Table 10; the ~2.5-day warning supports local spare reservation and scheduling, not overseas ordering.

5.5 Financial Impact and P90 Projection

Table 10 maps AI-recovered production into the existing P90 financial model, holding debt, tax, and non-AI baseline OPEX fixed. Avoided lead-time losses and AI software costs (excluded from Table 9; CAPEX/OPEX and net cash flow in footnote ¹) are the only varied terms. The five-month pilot estimate of 376.65 MWh is scaled to the P90 analysis horizon by $24 / 5.25 \approx 4.57$, yielding 1,721.52 MWh of modeled generation saved - a 24-month projection of the pilot recovery rate, not simple annualization (12/5). Under this mapping, project NPV rises from USD 8.517M to USD 8.581M (+USD 63.5k) and IRR from 14.73% to 14.77% (+0.04 pp). The project-level delta is small by design: readiness pays for itself and leaves bankability intact.

Table 10. Financial impact after mapping AI-data production into P90 financial model

Metric	Without AI	With AI	Delta
Project IRR	14.73%	14.77%	0.04%
NPV (USD'000)	8,517.36	8,580.83	63.47
Total AI revenue delta (USD'000)	-	141.94	141.94
Total generation saved (MWh)	-	1,721.52	1,721.52

AI software costs in the With-AI case: deployment CAPEX = USD 3.0k; software OPEX $\approx$ USD 5.38k over the analysis horizon (pilot service fee USD $1,176.47 \times 1,721.52/376.65$). Undiscounted net incremental cash flow $\approx$ USD 141.94k – 3.0k – 5.38k = USD 133.56k; the NPV delta of USD 63.5k is the discounted P90 value of this stream.

6 Ablation Study

We isolate each component’s contribution with a leave-one-out ablation over five variants, evaluating both event-centric detection quality and operational value.

Table 11. Component leave-one-out ablation.

Variant	Precision	Recall	F1-Score	Alert	NPV Gain (USD)
Threshold (Baseline)	0.0313	1.0000	0.0607	3226	0
w/o Peer-relative PR	0.0351	1.0000	0.0679	2874	6,560.26
w/o Adaptive Window	0.3077	0.9901	0.4695	325	51,121.55
w/o STL Decomposition	0.5470	0.9802	0.7021	181	51,121.55
Full Pipeline	0.5500	0.9802	0.7046	180	51,121.55

NPV gains in this ablation quantify each component's marginal contribution relative to the Threshold baseline over the [test-window / 52-candidate-event] set, and are not directly comparable to the full-horizon with-AI-versus-P90 delta of +USD 63.5k in Table 10.

Peer-relative PR is the most influential component: removing it reduces the NPV gain from USD 51,122 to USD 6,560 (−87.2%), because fleet-wide weather variation is then misattributed to inverter-specific loss. The adaptive window adds secondary responsiveness (−20.0%) and STL only stability (−2.9%). Chasing one metric is misleading. A static threshold hits recall 1.0000 but returns no operational value. What counts is the financial impact a component carries downstream, not its classification score on its own.

7 Discussion

Prior work addresses PV-maintenance components in isolation -Chinese grid-scale studies [11] deliver anomaly detection that ends at classification, while European frameworks [12], [18] embed financial modelling but assume near-certain spare-part availability, an assumption import-dependent Southeast Asian supply chains cannot make. This study's contribution is therefore not a new detector (F1 = 0.7046) but the composition of three absent layers: a label-free technical layer on standard 15-minute logs (STL improves fleet-PR reconstruction by 21.15%; achieving 98.02% recall and an F1 score of 0.7046 over 101 target events), an inventory layer (5 optimal spares; ~97% scenario-cost reduction versus reactive procurement), and a financial layer mapping avoided downtime into P90 projections -a combination made financially consequential by Decree 135/2024's shift to self-consumption, under which every downtime hour is uncompensated at EVN TOU tariffs up to 5,422 VND/kWh, and made investment-relevant by the DPPA framework (Decree 80/2024), where structured O&M evidence lowers the diligence barrier to FDI.

The framework transfers across rooftop and utility scale with parameter recalibration, as core inputs require no fault labels; however, validation on a single 26.15 MWp site leaves cross-site generalisation untested, the inventory advantage is horizon-dependent, and the human-in-the-loop reorder step remains untested in live operation, where user trust and human response latency could impact the effective 14-day lead time. The pilot deployment yielded an empirical generation recovery of 376.65 MWh over five months. Extrapolating this performance to a long-horizon projection, we estimate an horizon / P90 projection recovery of 1,721.52 MWh. While this linear extrapolation provides a baseline for the P90 assessment, future iterations must incorporate seasonal irradiance variations (e.g., monsoon effects) to yield a more rigorous annual model.

8 Conclusion and Future Work

This study extends commercial rooftop-PV inverter maintenance from a localized anomaly detection task into an integrated, techno-economic decision-support framework. Empirically validated on a 26.15 MWp Vietnamese fleet comprising 219 inverters and 93,277 inverter-days, the proposed zero-shot, unsupervised technical layer effectively extracts early risk signals from standard telemetry. Rigorously evaluated against supervised Gradient Boosting baselines across an extended 101-event suite, the pipeline achieves a 98.02% recall rate ($F1 = 0.7046$) and compared with the vanilla Isolation Forest baseline, the full pipeline reduces alert volume from 512 to 180 episodes while improving precision from 0.1953 to 0.5500. Under an asymmetric cost model, the zero-shot approach gives the lowest expected maintenance cost, USD 3,810 — a practical choice before enough labels exist to train supervised models.

Future work should expand empirical validation across heterogeneous environments, including utility-scale, floating, and mixed-brand PV fleets, while transitioning the batch-update pipeline to a real-time streaming architecture. With the progressive accumulation of ground-truth failure labels, subsequent iterations could employ supervised methodologies to calibrate heuristic triage scores into absolute predictive probabilities. Additionally, transfer learning paradigms could facilitate the adaptation of risk thresholds across novel climatic zones and inverter topologies. Finally, augmenting the financial module to model macroeconomic variables - such as exchange rates, hardware costs, and demand fluctuations - as probability distributions will yield a more rigorous probabilistic evaluation of IRR and Debt Service Coverage Ratio (DSCR) under systemic uncertainty.

9 References

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